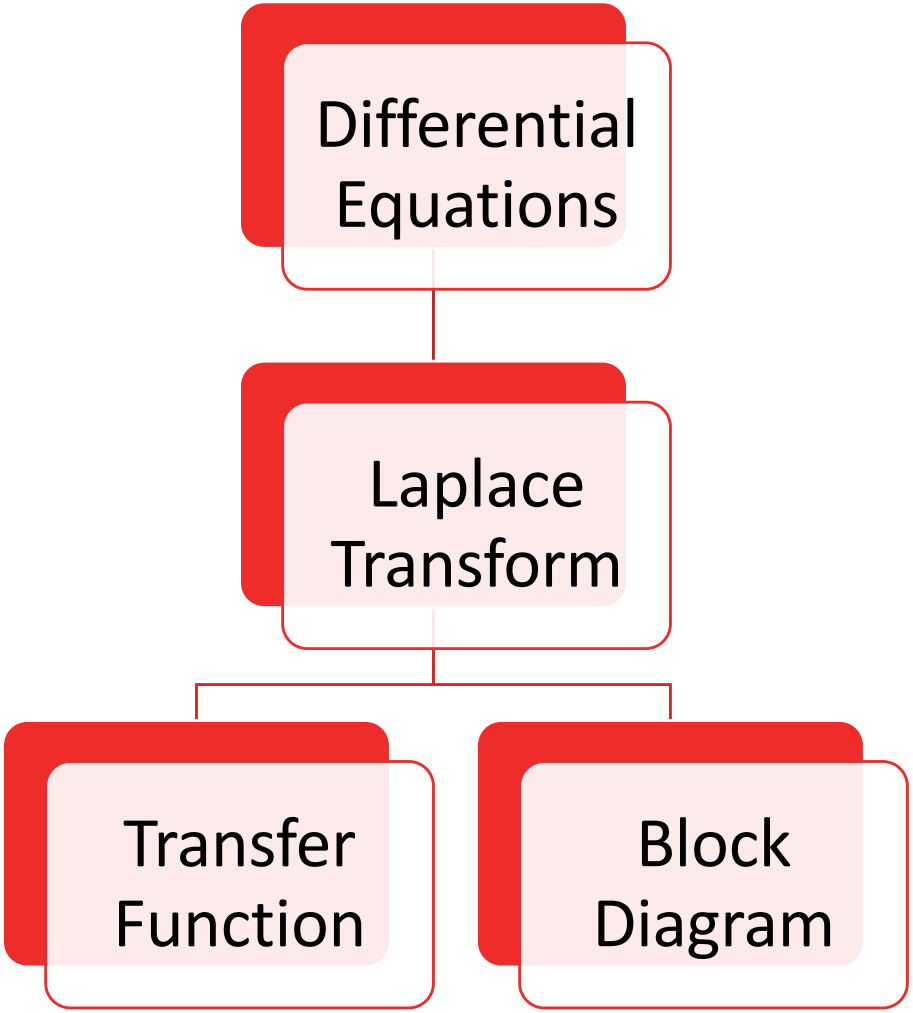
Mechatronics Design Laboratory I

ENME 561/661

Mahdis Bisheban (she/her), Ph.D., P.Eng
Associate Professor
Department of Mechanical and Manufacturing Engineering

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- Control
  - To cause an output variable (controlled variable) of a dynamic process to follow a desired reference variable accurately by applying the control signal to the plant
- Modelling
  - Developing a mathematical model
    - By using physical laws governing the system (e.g. Newton's laws, Kirchhoff's laws,..)
    - By doing system identification (examining the device, measuring its response to input, and constructing an analytical model from data)
    - A combination of the above
  - Usually, a Model is a set of differential equations that describes the dynamic behaviour accurately or fairly well.
    - We describe the physical world with differential equations


Differential Equations
Laplace Transform
Transfer Function
Block Diagram

Laplace Transform

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

Properties of Laplace Transform:

- $L[\,f(t)] = F(s)$
- Superposition
  - $L[a\,f(t) + b\,g(t)] = a\,F(s) + b\,G(s)$
  - $L[a\,f(t)] = a\,F(s)$
- Differentiation
  - $L\left[\frac{df}{dt}\right] = s\,F(s) - f(0^-)$
  - $L\left[\frac{d^2f}{dt^2}\right] = s^2\,F(s) - sf(0^-) - \frac{df}{dt}(0^-)$
  - ..
- Integration
  - $L\left[\int_0^t f(\zeta)d\zeta\right] = \frac{1}{s}F(s)$

Table of Laplace Transforms

Number	$F(s)$	$f(t), t \geq 0$
1	1	$\delta(t)$
2	$\frac{1}{s}$	$1(t)$
3	$\frac{1}{s^2}$	t
4	$\frac{2!}{s^3}$	t^2
5	$\frac{3!}{s^4}$	t^3
6	$\frac{m!}{s^{m+1}}$	t^m
7	$\frac{1}{(s+a)}$	e^{-at}
8	$\frac{1}{(s+a)^2}$	te^{-at}

Linear Time-Invariant (LTI) systems

- Linear systems
  - if the principle of superposition is applied

- Linear Differential Equation:
  - If the coefficients are constant or a function of independent variable (time)
- Linear Time-Invariant systems
  - Linear and coefficients are constant in time

Transfer Function

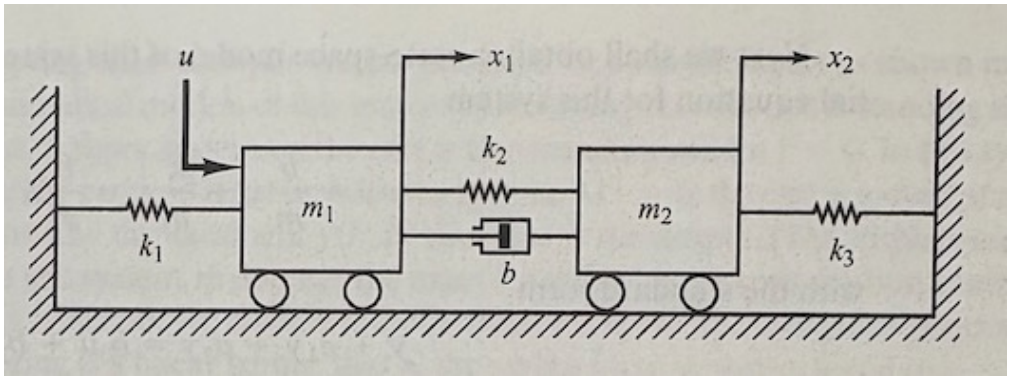
- To represent **LTI** differential equations (dynamic systems) by algebraic equations (in s-domain).
- To represent and study the Input and output relationship
  - Linear time-invariant differential equation (DE):

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1}\dot{y} + a_n y$$
$$= b_0 x^{(m)} + b_1 x^{(m-1)} + \cdots + b_{m-1}\dot{x} + b_m x \qquad (n \geq m)$$

  - Transfer function (Laplace transformed output to the Laplace transformed input with zero initial conditions):

$$\text{Transfer function} = G(s) = \left.\frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]}\right|_{\text{zero initial conditions}}$$
$$= \frac{Y(s)}{X(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1}s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n}$$

- Order of the system: the highest power of s in the denominator of the transfer function (the order of the DE)



Coupled Differential Equations:

$$m_1\ddot{x}_1 + b\dot{x}_1 + (k_1 + k_2)x_1 = b\dot{x}_2 + k_2x_2 + u$$

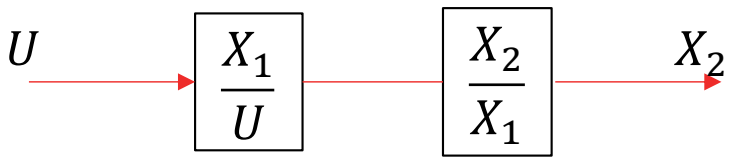
$$m_2\ddot{x}_2 + b\dot{x}_2 + (k_2 + k_3)x_2 = b\dot{x}_1 + k_2x_1$$

Transfer Functions (Laplace Transformed):

$$\frac{X_1(s)}{U(s)} = \frac{m_2s^2 + bs + k_2 + k_3}{(m_1s^2 + bs + k_1 + k_2)(m_2s^2 + bs + k_2 + k_3) - (bs + k_2)^2}$$

$$\frac{X_2(s)}{U(s)} = \frac{bs + k_2}{(m_1s^2 + bs + k_1 + k_2)(m_2s^2 + bs + k_2 + k_3) - (bs + k_2)^2}$$

Block Diagram:



Coupled differential equations:

$$m_1\ddot{x}_1 + b\dot{x}_1 + (k_1 + k_2)x_1 = b\dot{x}_2 + k_2x_2 + u$$

$$m_2\ddot{x}_2 + b\dot{x}_2 + (k_2 + k_3)x_2 = b\dot{x}_1 + k_2x_1$$

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Transfer Function (TF)

- S and TF(s) are in the complex domain

- Poles
- Zeros
- Gain

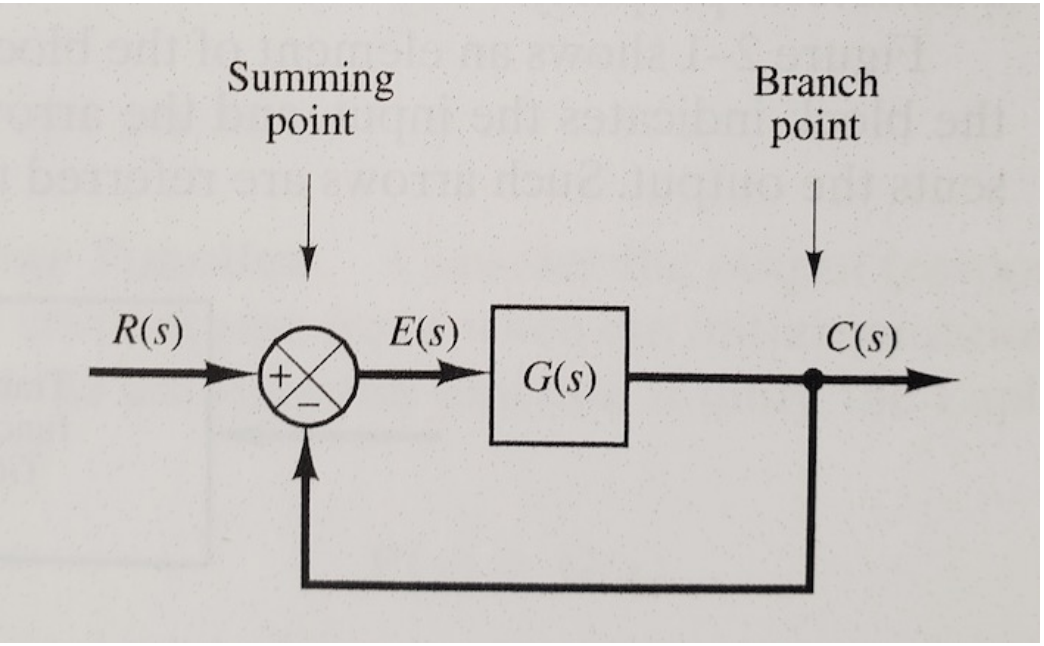
Example: $\frac{s+4}{(s+3)(s+6)}$

Application of TF

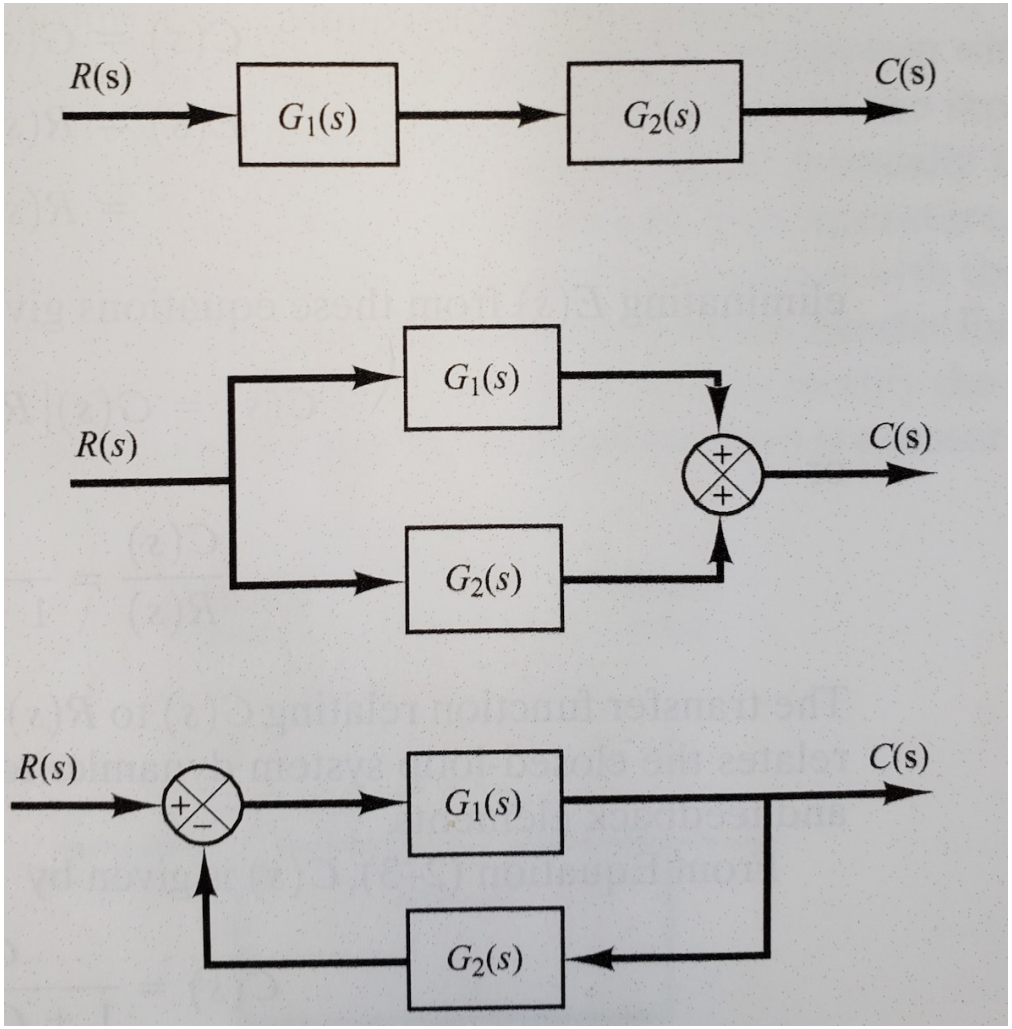
- Stability analysis
- Design of filters, controllers, …
- Analysis using Root locus, Bode plots, …

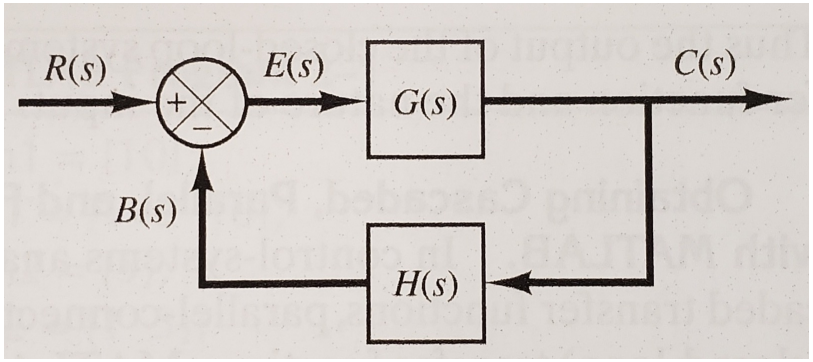
Block Diagram

- A visual representation of the functions performed by different components and of the flow of signals
- A **block** is a symbol for a mathematical operation on the input signal
- A **signal** can pass only in the direction of the arrow



Series, Parallel, and Feedback Transfer Functions





- Open Loop TF

$$\frac{B(s)}{E(s)}$$

- Feedforward TF

$$\frac{C(s)}{E(s)}$$

- Closed Loop TF

$$\frac{C(s)}{R(s)}$$

Reference

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